

Geometry Metric Curvature – Quantum Frequency Equivalence Principle: GR Metric- QFT Frequency Function($\omega_{\text{QFT}} = f(g_{\mu\nu})_{\text{GR}}$)

--Section 13 of the Quantized Elastic Cosmic Model, <https://doi.org/10.5281/zenodo.14788393>

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Abstract:

This paper proposes a novel method based on the Space Metric Deformation – Quantum Frequency Equivalence Principle, aiming to provide an operational mathematical tool for connecting General Relativity with Quantum Field Theory. The core idea of this principle is that any change in the metric of space (i.e., changes in spatial curvature) modulates the resonant frequency of the fundamental units composing space. As a result, the geometric properties of space can be characterized through the resonance frequencies of SEQ, thereby mapping changes in the space metric into the frequency domain. This mapping offers a new perspective on the relationship between gravity and quantum phenomena. In this model, electromagnetic wave frequency is the excited resonance frequency of SEQ along the wave's propagation path, while gravitational redshift serves as the direct experimental verification of this principle. This approach not only reveals the intrinsic link between spatial elastic response and quantum frequency but also lays a new mathematical foundation for unifying gravitational and quantum theories.

Introduction:

In General Relativity, the bending of the space metric is understood as the geometric manifestation of gravity, while Quantum Field Theory describes the quantum behavior of particles in flat spacetime. Both theories have achieved great success within their respective domains, yet their unification remains one of the greatest challenges in modern physics. The method proposed in this chapter, based on the Space Metric Deformation – Quantum Frequency Equivalence Principle, presents a new angle to bridge this theoretical gap.

Origin of the Principle: Gravitational Redshift Phenomenon:

Gravitational redshift is a classic prediction of General Relativity: when light propagates from a region of strong gravitational field to a weaker one, its frequency decreases. This phenomenon indicates that changes in the space metric (i.e., variations in gravitational potential) influence the frequency of electromagnetic waves. In the model proposed here, the frequency of an electromagnetic wave is no longer merely viewed as a solution to the wave equation, but rather as the resonance frequency of Space Elementary Quanta (SEQ) experienced along the wave's propagation path. This reinterpretation of frequency directly links the variation of the space metric to observable frequency shifts, making gravitational redshift a natural consequence of this principle.

Rethinking Space and Time from the Frequency Domain:

In this model, time is also redefined as the counting process of transformations within the

SEQ network, and frequency variations reflect the local evolution rate of spacetime. Therefore, frequency becomes not only a property of electromagnetic waves but also a manifestation of the dynamic characteristics of space itself. This perspective provides a unified framework for understanding relativistic effects such as time dilation and gravitational redshift, and opens up new avenues for the development of quantum gravity theory.

Bridging General Relativity and Quantum Field Theory:

One of the key advantages of this method is that it offers a mechanism to link the spatial curvature of General Relativity with the frequency concepts in Quantum Field Theory. By modulating the resonance frequencies of SEQ, the geometric properties of space can be encoded into the observable quantities of quantum systems, thereby providing an operational tool for constructing a unified theory of gravity and quantum mechanics. Moreover, the model can explain classical quantum phenomena such as the constancy of the speed of light, the uncertainty principle, and the double-slit experiment, while offering theoretical foundations for future experimental verification (e.g., measuring differences in magnetic moments between positrons and electrons).

Section 13:

13. Space Deformation - SEQ Resonant Frequency Modulation Duality

In Section 4, we discussed the action and mentioned that spatial deformation corresponds to frequency modulation. In this chapter, we will further explore this topic. Furthermore, the subsequent mathematical formalization of this model does not require a specific characterization of spatial deformation. Instead, it maps spatial deformation onto the corresponding region's frequency domain of SEQ network while distinguishing between compression phases, stretching phases, and twistors. Frequency modulation is used to characterize spatial deformation and potential energy changes.

13.1 Frequency Modulation as an Essential Description of Spatial Deformation

The model suggests that any metric change in space, such as curvature caused by gravitational fields, modulates the resonant frequency of SEQ. Compression and stretching phases influence frequency domain modulation through asymmetric elastic coefficients. This frequency modulation directly encodes the geometric information of spatial deformation, eliminating the need for additional Riemann geometry descriptions.

The traditional concept of potential energy terms (gravitational, electromagnetic, or quantum field potentials) is reinterpreted as frequency modulation of SEQ resonance. For instance, a decrease in gravitational potential energy corresponds to a frequency domain offset, while the release of potential energy manifests as dynamic modulation restoring the frequency to its high-frequency ground state. This mapping enables a unified frequency-domain representation of the metric field in general relativity and potential energy terms in quantum field theory.

Entropy Increase Rate: In addition, since the conduction frequency within a local space directly determines the local entropy increase rate of the system, there also exists a dualistic modulation mechanism between space geometry deformation and the rate of entropy increase. This relationship is self-consistent and analytically derivable under the SEQ quantized space

model.

Simplified Pathway for Mathematical Formalization:

13.2 Classification of Spatial Deformations: First, classify spatial deformations into 4 types (specific classifications can be refined based on future research):

Stretching Phase

Compression Phase

Left-handed Twistor

Right-handed Twistor

Since the model assumes that SEQ have a fixed chirality spin in their ground state, the frequency modulation caused by left-handed and right-handed twistors is not entirely symmetric. Based on QCD and cosmological observations, the elastic coefficients and frequency modulation of space should be nonlinear functions. Therefore, according to existing QCD, electromagnetism, and cosmological observations, preliminary modeling of the operator functions for these 4 deformations to SEQ frequency modulation can be established. Embedding these operators into the action function or other equations allows the analytic expression of action from Chapter 4 to represent the modulation of transmission frequency due to deformation.

13.3 Construction of Discrete Functional Framework:

By defining the local frequency response function of the SEQ network, continuous spatial deformation can be transformed into a parametric problem on discrete frequency lattice points. This model converts the geometrical dynamics of spatial tensor-twistor deformation into frequency dynamics, providing a new mathematical framework for unifying gravity and quantum theory. Its formal simplicity may offer new tools for physics. Future research should focus on developing specific algorithms for discrete functional equations and establishing mappings with parameters of the Standard Model. However, it should be noted that the specific mathematical modeling of this discrete model depends on the adjacency topology of the space SEQ network, which remains undetermined at present. Therefore, the mathematical phenomenological fitting is only approximate.

13.4 Generalized Coordinates in This Model:

Preliminary Discussion on the Lagrangian and Hamiltonian Analytical Mechanics Framework Based on This Model:

In the topologically homeomorphic space field of this quantized elastic spacetime model, each SEQ (Space Elementary Quantum) has a fixed spatial coordinate and fixed adjacency relations. The generalized coordinates of each SEQ are its resonance frequency and resonance axis vector. These two quantities, combined with the dynamic structural matrix of matter, can generate physical quantities such as mass, momentum, velocity, acceleration, force, kinetic energy, and potential energy—all expressed as polynomials of the generalized coordinates or their differentials (partial derivatives or integrals). Based on this framework, conservation equations can be constructed.

13.5 Under the topologically homeomorphic setting, the spatial coordinates of each SEQ

serve as its structural label and constitute important topological invariants. These coordinates remain fixed during dynamical evolution (i.e., they do not change over time), and therefore are not subjected to time differentiation. Nevertheless, they play a crucial role in constructing conservation equations, such as defining local gradients and adjacency relations. Hence, they should be regarded as background structural parameters rather than components of generalized coordinates.

13.6 Mass, Mass represents a spatial compression state that cannot be characterized by a single SEQ. Instead, it requires a local description in terms of the local spatial compression rate — the degree of deformation, which can be quantitatively expressed as the average frequency shift within that local space. Frequency directly reflects the extent of spatial deformation.

Since this model does not incorporate the conventional concept of volume, the notion of compression is inherently tied to the topologically homeomorphic SEQ network. Whether compressed or not, the structure remains a SEQ grid, and thus, spatial compression must be described through either:

A general relativistic tensor representation of the compression rate, or

The average resonance frequency of the SEQs contained within the mass-bearing object, which captures the spatial deformation effect.

Therefore, mass can be effectively represented by the number of SEQs within the object and their collective resonance frequency. This leads to the following formulation:

$$m = K \times N \times (\omega_p / \bar{\omega})$$

Where:

m: mass

K: a dimensional conversion constant (can be dimensionless or carry traditional mass dimensions)

N: number of SEQs contained in the mass-bearing object

$\bar{\omega}$: average resonance frequency (relative to Planck frequency shift)

ω_p : Planck frequency

$(\omega_p / \bar{\omega})$: represents the degree of spatial deformation

13.7 Force — next, we discuss $F = ma$. Acceleration can be understood as the rate of change of spatial deformation gradient between a mass-bearing object and its external environment. This gradient of deformation induces the object to maintain structural synchronization during acceleration, resulting in additional spatial deformation — compression at the front end of acceleration and stretching at the back end. In this framework, m represents the structural response barrier (mass), and F is the deformation action required to induce an additional gradient of spatial deformation in an object of mass m. The essence of $F = ma$ is: in order to cause a change in the spatial deformation gradient (acceleration) of an object that possesses a structural response barrier (mass), a corresponding deformation action (force) must be applied.

13.8 Other conceptual constructions of classical physical quantities — the following are only examples and not strict mathematical formalizations:

Starting from the perspective of spatial deformation — local frequency duality, we define the phase $\varphi = \Delta\omega / \omega_0$, where $\Delta\omega$ is the local frequency shift and ω_0 is the reference frequency. From this, we construct the wave vector $k = \nabla\varphi$. Then, we define momentum of generalized coordinate-SEQ Frequency as $p = \hbar k$, where $\hbar$ is the reduced Planck constant, $\hbar = h / (2\pi)$.

The resonance frequency and resonance axis vector of each SEQ — which are the generalized coordinates — correspond to two different sets of analytical expressions for generalized momentum. The generalized momentum of the SEQ resonance axis needs to be further developed in future work. According to this model, the direction of the SEQ resonance axis depends on the adjacency configuration, which is discrete and finite, thus making the direction quantized as well.

In this framework, the reduced Planck constant $\hbar$ is not used to characterize quantization of angular momentum, but rather to describe phase quantization in the harmonic oscillation of SEQ (this is only a conceptual analysis tool — the actual phase variation may not necessarily be quantized). This allows for a formal description of the phase gradient between matter and its surrounding SEQ in the process of material wave formation.

Velocity is then defined as $v = p / m$, acceleration as $a = dv / dt$, force as $F = dp / dt$, and so on.

13.9 In this model, the spatial field is a topologically homeomorphic structural field, and the motion of matter corresponds to the propagation of excitation waves on this field. Therefore, when constructing conservation equations, they may differ from the traditional conservation equations in analytical mechanics, but they should be able to derive equivalent forms corresponding to the classical ones. Here, I can only propose a rough framework, and the specific rigorous mathematical formalization likely still requires the work of professional physicists to complete.

Although Einstein adopted the geometric description based on the metric tensor in his General Relativity, he had already noticed the significant influence of gravity on light frequency shifts. The 'Deformation-Frequency Equivalence Principle' proposed in this paper can be regarded as a continuation of this unification vision—it reinterprets the variation of spacetime tensor structure as a frequency modulation process of SEQ, thereby establishing a new unified descriptive paradigm between the microscopic and macroscopic scales.

Deriving a metric-quantum frequency function from cosmological gravitational redshift measurements and QCD-level interactions could establish a functional mapping($\omega_{\text{QFT}} = f(g_{\mu\nu})_{\text{GR}}$) between GR's spacetime geometry and QFT's quantum observables, offering a mathematically tractable and empirically testable tool for unifying gravity and quantum theory.